甲、填充題：共 10 題，每題 8 分，共 80 分。請在答案卷上列出題號依序作答。

請注意：本（甲）部分，共 10 題，命題型態為填充題，請依題號順序獨立列出，勿同時陳列出計算過程。倘若答案被包含在演算過程，將被視為試算流程，不另行挑出計分。

1. Evaluate the limit $\displaystyle\lim_{x\to 0}\frac{\sqrt{1+\tan 3x}-\sqrt{1+\sin 3x}}{x^3}$.

2. If $f(x)=\displaystyle\int_0^{g(x)}\frac{1}{\sqrt{1+t^3}}\,dt$, where $g(x)=\displaystyle\int_0^{\cos 2x}\left[1+\sin\left(t^2\right)\right]dt$, find $f'(\pi/4)$.

3. Find the volume of the solid obtained by rotating about the $x$-axis the region enclosed by the curves $y=\dfrac{4}{x^2+4}$, $y=0$, $x=0$, and $x=2$.

4. Find the absolute maximum value of $f(x,y)=e^{-x^2-y^2}\left(x^2+2y^2\right)$ on the set $D$, where $D$ is the disk $x^2+y^2\le 4$.

5. Suppose $z=f(x,y)$, where $x=g(s,t)$, $y=h(s,t)$, $g(1,2)=3$, $g_s(1,2)=-1$, $g_t(1,2)=4$, $h(1,2)=6$, $h_s(1,2)=-5$, $h_t(1,2)=10$, $f_x(3,6)=7$, and $f_y(3,6)=8$. Find $\partial z/\partial s$ when $s=1$ and $t=2$.

6. Find the directional derivative of the function $f(x,y)=\dfrac{x}{x^2+y^2}$ at the point $P(1,2)$ in the direction of $\mathbf{v}=\langle 3,4\rangle$.

7. Evaluate the integral $\displaystyle\iint_R\left(1+\frac{y}{x}\right)dA$, where $R$ is the region enclosed by the lines $x+y=1$, $x+y=3$, $y=2x$, and $y=x/2$.

8. Find the average value of the function $f(x)=\displaystyle\int_x^2\cos\left(t^2\right)dt$ on the interval $[0,2]$.

9. Evaluate the integral $\displaystyle\int_{-2}^{2}\int_0^{\sqrt{4-y^2}}\int_{-\sqrt{4-x^2-y^2}}^{\sqrt{4-x^2-y^2}} y^2\sqrt{x^2+y^2+z^2}\,dz\,dx\,dy$.

10. Evaluate $\displaystyle\int_C\mathbf{F}\cdot d\mathbf{r}$, where $\mathbf{F}(x,y)=\langle\sqrt{x^2+1},\tan^{-1}x\rangle$ and $C$ is the triangle from $(0,0)$ to $(1,1)$ to $(0,1)$ to $(0,0)$.

注意：背面有試題

乙、計算、證明題：共 2 題，每題 10 分，共 20 分。須詳細寫出計算及證明過程，否則不予計分。

1. **(a)** (5 分) Use the **integral test** to determine whether the series $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\ln n}}$ is convergent or divergent.

   **(b)** (5 分) Find the interval of convergence of the series $\sum_{n=1}^{\infty} \frac{(x+2)^n}{n4^n}$.

2. **(a)** (5 分) Evaluate the integral $\int_1^{\infty} \frac{1}{x+x^3}\,dx$.

   **(b)** (5 分) Evaluate the integral $\int_1^{\infty} \frac{\tan^{-1} x}{x^2}\,dx$.

注意：背面有試題